

Q16. Arrays

— Not Submitted —

Q17. Arrays

— Not Submitted —

Q18. Arrays

— Not Submitted —

Q19. Arrays

— Not Submitted —

Q20. Arrays

— Not Submitted —

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Q1. Arrays

CODE

```
#include <stdio.h>

int main() {
    int production[12];
    int target = 5000;
    int annualProduction = 0;
    float averageProduction, efficiency;
    int validMonths = 0;
    char *monthNames[12] = {"January", "February", "March", "April", "May", "June",
                           "July", "August", "September", "October", "November", "December"};
    int shutdownFlag = 0;

    printf("Enter monthly production for 12 months (Jan-Dec).\n");
    printf("Enter -1 for invalid data, -999 for equipment failure (plant shutdown):\n\n");

    for (int i = 0; i < 12; i++) {
        printf("Enter production for %s: ", monthNames[i]);
        scanf("%d", &production[i]);

        / Break if equipment failure occurs
        if (production[i] == -999) {
            printf("Equipment failure reported in %s. Plant shutting down.\n", monthNames[i]);
            shutdownFlag = 1;
            break;
        }

        / Skip invalid production values
        if (production[i] < 0) {
            printf("Invalid production value for %s. Skipping this month.\n", monthNames[i]);
            continue;
        }

        / Process valid month
        annualProduction += production[i];
        validMonths++;

        / Determine incentive and inspection status for the month
        if (production[i] > target) {
            float incentive = production[i] * 0.10;
            printf("%s: Production = %d | Target Exceeded! Incentive = %.2f units\n",
                  monthNames[i], production[i], incentive);
        } else if (production[i] < (0.8 * target)) {
            printf("%s: Production = %d | Below 80% of target! Marked for INSPECTION.\n",
                  monthNames[i], production[i]);
        } else {
            printf("%s: Production = %d | Within acceptable range.\n",
```

```

        monthNames[i], production[i]);
    }
}

printf("\n- ANNUAL SUMMARY - \n");

if (shutdownFlag) {
    printf("Note: Plant shut down early due to equipment failure.\n");
}

if (validMonths > 0) {
    averageProduction = (float) annualProduction / validMonths;
    efficiency = (averageProduction / target) * 100;

    printf("Total Valid Months Processed : %d\n", validMonths);
    printf("Annual Production : %d batteries\n", annualProduction);
    printf("Average Monthly Production : %.2f batteries\n", averageProduction);
    printf("Production Efficiency : %.2f% \n", efficiency);

    / Overall decision-making
    if (efficiency > 100)
        printf("Overall Status: Excellent performance, incentive applicable.\n");
    else if (efficiency < 80)
        printf("Overall Status: Poor performance, inspection required.\n");
    else
        printf("Overall Status: Satisfactory performance.\n");
} else {
    printf("No valid production data available.\n");
}

return 0;
}

```

INPUT 5500

```

6200
-1
4200
5000
5100
5300
4800
-1
6100
4000
-999

```

OUTPUT (no output)

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Q2. Arrays

CODE

```

#include <stdio.h>

int main() {
    int i;
    int projectCount = 20;

    int quality[20];
    int delay[20];
    int cancelled[20];

    double qualitySum = 0.0;
    int evaluatedCount = 0;
    int excellentCount = 0, goodCount = 0, averageCount = 0, needsCount = 0;

    for (i = 0; i < projectCount; i++) {
        / Input format for each project:
        / cancelled(0/1) quality(0-100) delay(days)
        scanf("%d %d %d", &cancelled[i], &quality[i], &delay[i]);

        if (cancelled[i] == 1) {
            continue; / ignore cancelled projects
        }

        evaluatedCount++;
        qualitySum += quality[i];
    }
}

```

```

/ Classification based on criteria
if (quality[i] > 90 & delay[i] = 2) {
    excellentCount+ ;
} else if (quality[i] > 75 & quality[i] < 89 & delay[i] = 5) {
    goodCount+ ;
} else if (quality[i] > 60 & quality[i] < 74) {
    averageCount+ ;
} else {
    needsCount+ ;
}

/ break if management stops evaluation (cancelled input -1 as stop)
/ (If your faculty uses different rule, tell me the exact statement.)
if (cancelled[i] = -1) {
    break;
}

}

double avgQuality = (evaluatedCount > 0) ? (qualitySum / evaluatedCount) : 0.0;
double deliveryPercentage = (evaluatedCount > 0)
    ? (((double)excellentCount + goodCount) / evaluatedCount) * 100.0
    : 0.0;

printf("Average Quality = %.2f\n", avgQuality);
printf("Delivery Percentage = %.2f\n", deliveryPercentage);

printf("Excellent = %d\n", excellentCount);
printf("Good = %d\n", goodCount);
printf("Average = %d\n", averageCount);
printf("Needs Improvement = %d\n", needsCount);

return 0;
}

```

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INPUT₁

```

85
5
1
1

2
92
2
1
1

```

OUTPUT (no output)

Q3. Arrays

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Q4. Arrays

— Not Submitted —

Q5. Arrays

— Not Submitted —

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Q6. Arrays

CODE

```

#include <stdio.h>

int main() {
    const int TOTAL_CHIPS = 10000;
    double defectPercentages[30];
    char grades[30];

    int day;
    for (day = 0; day < 30; day+ ) {
        int status;
        int defectiveChips;
    }
}

```

```

/ status: 1 = normal, 2 = maintenance, 3 = breakdown
printf("\nDay %d\n", day + 1);
printf("Enter status (1=normal, 2=maintenance, 3=breakdown): ");
scanf("%d", &status);

if (status = 2) {
    / maintenance day: skip processing
    continue;
}

if (status = 3) {
    / breakdown: stop processing further days
    break;
}

printf("Enter defective chips for Day %d: ", day + 1);
scanf("%d", &defectiveChips);

defectPercentages[day] = (defectiveChips * 100.0) / TOTAL_CHIPS;

if (defectPercentages[day] < 1.0)
    grades[day] = 'A';
else if (defectPercentages[day] > 1.0 & defectPercentages[day] < 3.0)
    grades[day] = 'B';
else if (defectPercentages[day] > 3.0 & defectPercentages[day] < 5.0)
    grades[day] = 'C';
else
    grades[day] = 'R'; / Reject
}

printf("\n- Results for processed days - \n");
for (day = 0; day < 30; day+ ) {
    / If it was maintenance, we never filled grade value.
    / So show only valid days where grade is set.
    if (grades[day] != 'A' & grades[day] != 'B' & grades[day] != 'C' & grades[day] != 'R')
        continue;

    int accepted = TOTAL_CHIPS - (int)((defectPercentages[day] / 100.0) * TOTAL_CHIPS);

    printf("Day %d: Defect = %.2f% , Grade = %c, Accepted Chips = %d\n",
        day + 1, defectPercentages[day], grades[day], accepted);
}

return 0;
}

```

INPUT₁

```

50
2
1
120
1
250
1
400
1
600
3

```

OUTPUT (no output)

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Q7. Arrays

CODE

```

#include <stdio.h>

int main() {
    int n, i;
    int active;
    int planType;
    int usedGB;
    int remaining;
    int totalCapacity;
    int totalExtraCharges;
    int totalExtraGB;
}

```

```

int serverUsed;

printf("Enter number of customers: ");
if (scanf("%d", &n) != 1) return 0;

/ Server storage capacity for this scenario (max storage allowed) *
serverUsed = 0;

for (i = 0; i < n; i++) {
    printf("\nCustomer %d\n", i + 1);

    printf("Enter active status (1=active, 0=inactive): ");
    scanf("%d", &active);

    / continue for inactive accounts *
    if (active == 0) {
        printf("Inactive account -> skipped using continue.\n");
        continue;
    }

    printf("Enter plan type (1=Basic 2=Standard 3=Enterprise): ");
    scanf("%d", &planType);

    printf("Enter used storage in GB: ");
    scanf("%d", &usedGB);

    / Determine total capacity based on plan *
    if (planType == 1) totalCapacity = 100;
    else if (planType == 2) totalCapacity = 500;
    else totalCapacity = 2048; / Enterprise ~ 2 TB *

    / If server storage becomes full -> break *
    / We'll treat "server full" as total used across processed active customers exceeding a server limit. For simplicity and safety: server limit
    equals Enterprise capacity (2048 GB). *
    if (serverUsed + usedGB > 2048) {
        printf("Server storage is full -> stopping processing using break.\n");
        break;
    }

    serverUsed = serverUsed + usedGB;

    / Calculate remaining storage and extra charges *
    remaining = totalCapacity - usedGB;

    totalExtraGB = 0;
    totalExtraCharges = 0;

    if (remaining < 0) {
        totalExtraGB = -remaining; / extra GB *
        totalExtraCharges = totalExtraGB * 4; / ?4 per GB *
        remaining = 0; / no remaining if exceeded plan *
    }

    / Decision for warning if usage exceeds 95% *
    printf("Plan capacity: %d GB\n", totalCapacity);
    printf("Used: %d GB\n", usedGB);
    printf("Remaining (within plan): %d GB\n", remaining);
    printf("Extra charges: %d (Extra GB: %d)\n", totalExtraCharges, totalExtraGB);

    if (usedGB > (95 * totalCapacity) / 100) {
        printf("Warning: Storage usage exceeded 95% .\n");
    } else {
        printf("Storage status: Normal (< 95% ).\n");
    }
}

printf("\nTotal server used (processed customers): %d GB\n", serverUsed);
return 0;
}

```

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INPUT 4

```

1
3
900
1
3

```

900
1
1
1
50

OUTPUT (no output)

Q8. Arrays

— Not Submitted —

Q9. Arrays

— Not Submitted —

Q10. Arrays

— Not Submitted —

Q11. Arrays

— Not Submitted —

Q12. Arrays

— Not Submitted —

Q13. Arrays

— Not Submitted —

Q14. Arrays

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Q15. Arrays

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Q1. Arrays

CODE

```
#include <stdio.h>
```

```
#define MONTHS 12
```